Thursday 12 November | 15:30–16:00

A topos for étale-finite Heyting algebras

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Which lattices can arise as lattices of sentences, modulo provable equivalence, of intuitionistic higher-order theories? Categorically, this asks which Heyting algebras are *topos-admissible*: isomorphic to the lattice of subobjects of the terminal object of an elementary topos, also called its lattice of truth values. The question whether every Heyting algebra is topos-admissible appeared in print in a 1992 paper by Pitts [1], but dates back at least to 1980, when Pitts learned it from Freyd.

Recently, Xu and Ye answered this question negatively, showing that the free Heyting algebra on two generators is not topos-admissible [2]. This leaves open the problem of finding an intrinsic characterisation of the Heyting algebras that can occur as lattices of truth values.

Known positive cases include complete Heyting algebras (frames), via sheaf toposes, and Boolean algebras, via a construction due to Freyd. In joint work with Rodrigo Almeida and Igor Arrieta [3], we extend Freyd's construction to all *étale-finite Heyting algebras*, a class introduced by Kuznetsov. These algebras satisfy a generalisation of the law of excluded middle relative to a finite Heyting subalgebra, and include examples that are neither Boolean nor complete. Using Esakia duality, we construct, for each such algebra, an elementary topos of certain compact étale spaces whose lattice of truth values is isomorphic to the given algebra.

Keywords: Elementary toposes; Heyting algebras; Esakia duality; étale-finite algebras
Subject area: Categorical logic and topos theory
MSC2020: 03G30 (primary); 18B25, 06D20, 06D50 (secondary)

References

[1] A. M. Pitts, *On an interpretation of second order quantification in first order intuitionistic propositional logic*, Journal of Symbolic Logic **57**(1), 33–52, 1992.

[2] L. Ye and Y. Xu, *Failure of higher-order truth within intuitionistic propositional logic*, preprint, 2026. arXiv:2608.26874.

[3] M. Abbadini, R. N. Almeida and I. Arrieta, *A topos for étale-finite Heyting algebras*, preprint, 2026. arXiv:2606.03861.